

# A canonical-weight phase diagram for the Schatten Berger–Coburn problem

Candidate partial answer to Open Problem 1 in arXiv:2312.06656

12 August 2026

**Status: candidate partial result.** For the canonical doubling Fock weights  $\phi_m(z) = |z|^m$ , the Berger–Coburn question is answered for every  $1 < p < \infty$  when  $m \leq 2$ , and for all but one high- $p$  range when  $m > 2$ . The low endpoint is sharp: the positive Muckenhoupt window meets an explicit counterexample window exactly, with logarithmic failure at the boundary.

## 1 Question and phase diagram

Let  $F_m^2 = F_{|z|^m}^2$  and let  $H_f : F_m^2 \rightarrow L_{|z|^m}^2$  be the Hankel operator with bounded symbol  $f$ . Asghari, Hu, and Virtanen [1] prove the Berger–Coburn phenomenon for the Hilbert–Schmidt class and ask whether

$$H_f \in \mathcal{S}_p \iff H_{\bar{f}} \in \mathcal{S}_p \quad (1 < p < \infty)$$

on an arbitrary doubling Fock space. Their Remark 6 identifies  $\rho^{p-2} \in A_p$  as the one missing ingredient in their proof strategy, where  $\rho(z)$  is the radius of a disk of unit  $\Delta\phi$ -measure.

For  $m > 2$  set

$$p_-(m) = \frac{2m}{m+2}, \quad p_+(m) = \frac{2m}{m-2}.$$

**Theorem 1** (Canonical-weight phase diagram). *Let  $m > 0$ ,  $1 < p < \infty$ , and  $f \in L^\infty(\mathbb{C})$ .*

1. *If  $0 < m \leq 2$ , then  $H_f \in \mathcal{S}_p$  if and only if  $H_{\bar{f}} \in \mathcal{S}_p$ , and their Schatten norms are comparable.*
2. *If  $m > 2$  and  $p_-(m) < p < p_+(m)$ , the same conclusion holds.*
3. *If  $m > 2$  and  $1 < p \leq p_-(m)$ , the bounded Xia symbol*

$$f(z) = \begin{cases} z^{-1}, & |z| \geq 1, \\ 0, & |z| < 1, \end{cases} \tag{1}$$

*satisfies  $H_f \in \mathcal{S}_p$  but  $H_{\bar{f}} \notin \mathcal{S}_p$ . At  $p = p_-(m)$  the obstructing integral diverges logarithmically.*

*Thus, for  $m > 2$ , only  $p \geq p_+(m)$  remains undecided.*

The two endpoints are Hölder conjugates:  $p_-^{-1} + p_+^{-1} = 1$ . This symmetry is suggestive, but it does not by itself transfer either Schatten membership or nonmembership between the two ranges.

## 2 The exact Muckenhoupt calculation

The source's canonical-weight lemma states that

$$\rho(z) \asymp |z|^{1-m/2} \quad (|z| \gg 1). \quad (2)$$

Since  $\rho$  is positive and comparable on compact sets, for every fixed  $p$  the weight  $\rho^{p-2}$  is comparable on all of  $\mathbb{C}$  to

$$w_\gamma(z) = (1 + |z|)^\gamma, \quad \gamma = (1 - m/2)(p - 2).$$

Membership in  $A_p$  is unchanged under two-sided comparability.

**Lemma 1** (Regularized planar powers). *For  $1 < p < \infty$ ,*

$$(1 + |z|)^\gamma \in A_p(\mathbb{C}) \iff -2 < \gamma < 2(p - 1).$$

*Proof.* Test the  $A_p$  expression on  $B(0, R)$  as  $R \rightarrow \infty$ . If  $\gamma \leq -2$ , the average of  $w_\gamma$  is respectively  $\asymp R^{-2} \log R$  or  $\asymp R^{-2}$ , while the dual average supplies the uncanceled positive power; hence the expression is unbounded. Applying the same argument to  $w_\gamma^{-1/(p-1)}$  gives necessity of  $\gamma < 2(p - 1)$ , including logarithmic failure at equality.

Conversely, assume the strict inequalities and let  $B = B(a, R)$ . If  $R \leq (1 + |a|)/2$ , both weights are essentially constant on  $B$ , so their normalized averages cancel. Otherwise  $|a| < 2R + 1$  and  $B$  lies in a centered disk of radius  $O(R + 1)$ . The conditions  $\gamma > -2$  and  $-\gamma/(p - 1) > -2$  give

$$\frac{1}{|B|} \int_B w_\gamma \lesssim (R + 1)^\gamma, \quad \frac{1}{|B|} \int_B w_\gamma^{-1/(p-1)} \lesssim (R + 1)^{-\gamma/(p-1)}.$$

Their  $A_p$  product is uniformly bounded. □

When  $0 < m < 2$ , put  $a = 1 - m/2 \in (0, 1)$ . Then  $\gamma = a(p - 2) > -2$  and  $\gamma < 2(p - 1)$  for every  $p > 1$ ; the case  $m = 2$  has  $\gamma = 0$ . When  $m > 2$ , put  $d = m/2 - 1 > 0$ , so  $\gamma = -d(p - 2)$ . The two strict inequalities in Lemma 1 become

$$p < 2 + \frac{2}{d} = \frac{2m}{m - 2} = p_+(m), \quad p > \frac{2d + 2}{d + 2} = \frac{2m}{m + 2} = p_-(m).$$

Consequently  $\rho^{p-2} \in A_p$  precisely in the positive ranges of Theorem 1. Remark 6 of the source, using the weighted Ahlfors–Beurling estimate of Dragičević [2], then gives

$$\|H_{\bar{f}}\|_{\mathcal{S}_p} \lesssim \|H_f\|_{\mathcal{S}_p}.$$

Apply the same statement to  $\bar{f}$  for the reverse implication and norm comparison. This proves parts (1) and (2).

## 3 The counterexample reaches the lower endpoint

The source proves much more about the one-sided membership of (1) than its stated  $p \leq 1$  theorem needs. Outside a sufficiently large disk,  $f$  is holomorphic on every intrinsic disk  $D(z, \rho(z))$ . Hence its integral distance to analytic functions vanishes there. On the remaining compact set it is bounded, and local finiteness of  $\rho^{-2}dA$  gives

$$f \in \text{IDA}^{p,2,-2/p} \quad \text{and therefore} \quad H_f \in \mathcal{S}_p \quad \text{for every } p > 0$$

by Theorem 1.2 of the source.

For  $|z|$  large, the direct mean-oscillation calculation in the same proof gives

$$MO_{2,1}(f)(z) \asymp \frac{\rho(z)}{|z|^2}. \quad (3)$$

Theorem 5.4 of the source says that simultaneous membership of  $H_f$  and  $H_{\bar{f}}$  in  $\mathcal{S}_p$  is equivalent to  $f \in \text{IMO}^{p,2,-2/p}$ . By (3), its tail is finite exactly when

$$\int_{|z|>R} \rho(z)^{-2} MO_{2,1}(f)(z)^p dA(z) \asymp \int_R^\infty r^{(1-m/2)(p-2)-2p+1} dr$$

is finite. For  $m > 2$ , the exponent is at least  $-1$  exactly when

$$p \leq \frac{2m}{m+2} = p_-(m).$$

At equality it is  $-1$ . Thus  $f \notin \text{IMO}^{p,2,-2/p}$  throughout this closed range. Since  $H_f \in \mathcal{S}_p$ , Theorem 5.4 forces  $H_{\bar{f}} \notin \mathcal{S}_p$ , proving part (3).

## 4 Scope and upgrade audit

The result is sharp at  $p_-(m)$  but does not settle  $p \geq p_+(m)$  for  $m > 2$ . Failure of the  $A_p$  sufficient condition is not itself a counterexample. Schatten nesting cannot propagate the Xia nonmembership upward, and interpolation supplies no endpoint estimate for a well-defined symbol-conjugation operator. Nor does unboundedness of the Beurling transform automatically produce a bounded symbol with the required IDA/IMO asymmetry.

The natural exterior-holomorphic variants also stop here: a bounded function holomorphic off a disk has a Laurent expansion whose slowest nonconstant decay is  $1/z$ , while  $1/z^n$  only weakens the tail for  $n > 1$ . Eight focused upgrade attempts, including these high- $p$  routes and a later-literature search through 12 August 2026, found no defensible extension. The general doubling-weight problem likewise remains open.

## References

- [1] G. Asghari, Z. Hu, and J. A. Virtanen, *Schatten class Hankel operators on doubling Fock spaces and the Berger–Coburn phenomenon*, arXiv:2312.06656v2 (2024); J. Math. Anal. Appl. 540 (2024), 128596.
- [2] O. Dragičević, *Weighted estimates for powers and the Ahlfors–Beurling operator*, Proc. Amer. Math. Soc. 139 (2011), 2113–2120.